Title

Titanic Survival Prediction Using Decision Tree Classifier

Objective

The objective of this project is to build a machine learning model that predicts whether a passenger survived the Titanic disaster based on features such as passenger class, gender, age, and fare.

Dataset

Dataset Name: Titanic Dataset

The dataset contains information about Titanic passengers, including:

- Passenger Class (Pclass)
- Gender (Sex)
- Age
- Fare
- Survival Status (Survived)

Tools and Technologies

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Jupyter Notebook / Google Colab

Methodology

Data Preprocessing

- Loaded the dataset using Pandas.

- Selected important features.

- Handled missing values in the Age column.

- Converted categorical data into numerical format.

Model Building

A Decision Tree Classifier was used to train the model on the dataset.

Model Evaluation

The dataset was divided into training and testing sets. Predictions were made on the test data and evaluated using:

- Accuracy Score

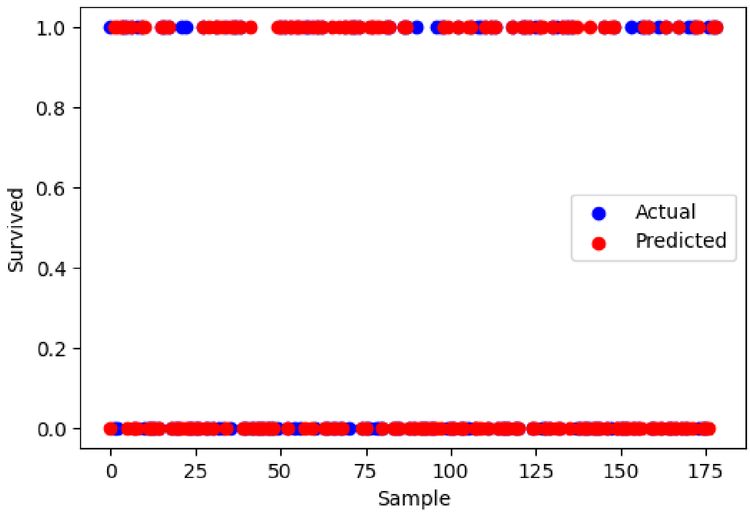
- Confusion Matrix

Results

The Decision Tree model successfully predicted passenger survival. The model achieved good accuracy and demonstrated the effectiveness of machine learning techniques for classification problems.

Visualization

The scatter plot compares actual and predicted survival outcomes. Most predicted values align with the actual values, indicating that the Decision Tree model is able to classify passenger survival with reasonable accuracy.



Conclusion

This project demonstrates the application of machine learning in predictive analytics. Using the Titanic dataset, a Decision Tree Classifier was trained and evaluated successfully. The model can be used to predict survival outcomes based on passenger information.

References

- Titanic Dataset

- Scikit-learn Documentation

- Pandas Documentation

- Matplotlib Documentation